# The Blueprint: Global Automated Mesh

A Complete Deployment Guide by Nathaniel Edet

## Phase 1: Local Environment Setup

Before running the automation, you need a local workspace and the necessary security credentials.

### Step 1: Generate SSH Keys

Open your terminal and create a dedicated key pair for this project. This allows Ansible to talk to your AWS instance securely.

```
ssh-keygen -t rsa -b 4096 -f ./final-key
```

Add **`final-key`** and **`final-key.pub`** to your **`.gitignore`** to keep them private.

## Phase 2: Infrastructure as Code (Terraform)

This creates the virtual "house" where your application will live.

### 1. terraform/main.tf

```
resource "aws_instance" "mesh_worker" {
  ami           = "ami-0c55b159cbfafe1f0" # Ubuntu 22.04
  instance_type = "t2.micro"
  key_name      = aws_key_pair.final-key.key_name
  vpc_security_group_ids = [aws_security_group.mesh_sg.id]
```

```
  tags = { Name = "mesh-worker-node" }
}

output "public_ip" {
  value = aws_instance.mesh_worker.public_ip
}
```

## Phase 3: Configuration Management (Ansible)

This installs the "furniture" (Docker and Kubernetes) inside your house.

### 1. ansible/inventory.ini

```
[webserver]
target_server ansible_host={{ TARGET_IP }} ansible_user=ubuntu
```

### 2. ansible/setup.yml (Partial)

```
- name: Setup Mesh Node
  hosts: webserver
  become: yes
  tasks:
    - name: Install K3s
      shell: curl -sfL https://get.k3s.io | sh -

    - name: Wait for K3s to initialize
      pause: seconds=15
```

## Phase 4: The CI/CD Pipeline (GitHub Actions)

This is the "brain" that connects everything. Save this as `.github/workflows/deploy.yml`.

```
name: "Global Mesh Deployment"
on: [push]
jobs:
  deploy:
    runs-on: ubuntu-latest
```

```
    steps:
      - uses: actions/checkout@v4
      - name: Terraform Init & Apply
        run: |
          cd terraform
          terraform init
          terraform apply -auto-approve
          echo "NEW_IP=$(terraform output -raw public_ip)" >> $GITHUB_ENV
      - name: Run Ansible
        run: |
          echo "${{ secrets.SSH_PRIVATE_KEY }}" > ./final-key
          chmod 400 ./final-key
          ansible-playbook -i ansible/inventory.ini ansible/setup.yml \
            --extra-vars "TARGET_IP=${{ env.NEW_IP }}" --private-key ./final-key
```

## Phase 5: Post-Deployment Manual Verification

Once the pipeline is green, SSH into your server to verify the Kubernetes pods.

```
ssh -i ./final-key ubuntu@<YOUR_IP>
sudo kubectl get nodes
sudo kubectl get pods
```

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